

Mack Thompson

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EDUCATION

Auburn University

Expected May 2027

B.S. in Computer Science & Mathematics, Minor in Business

GPA: 3.70/4.00

- Competitive Programming Team, Association for Computing Machinery (ACM), TFAS Charity Boxer, ΔXA fraternity

EXPERIENCE

Forward Deployed Engineer

Aug 2026 – Present

Harmony – AI operating system for manufacturing

Chattanooga, TN

- Deploying AI and software systems inside manufacturing facilities by integrating factory data, machines, and operational workflows.

Cloud Engineer Intern

May 2026 – Aug 2026

Hewlett Packard Enterprise / Juniper Networks

Sunnyvale, CA

- Built a Playwright crawler that maps 6,000+ UI elements and immediately shows how the application changes between software releases, where approximately 20% of the interface hierarchy shifts.
- Automatically updated the element paths used by test scripts, preventing approximately 20 broken tests per release and saving 1–2 days of QA developer work.
- Built the tool for planned rollout to additional HPE platforms, enabling teams to reuse the same automated release-validation process.

Software Engineer Intern

Sep 2025 – Mar 2026

OCV Apps – Public safety mobile apps

Opelika, AL

- Reduced management overhead by approximately 80% for 40+ apps by building a centralized configuration architecture that eliminated duplication across live manifests.
- Designed product-layer features including semantic search, AI-assisted navigation, and guided user journeys.

Research Assistant

Feb 2025 – Sep 2025

Auburn University – Computer Vision & Machine Learning

Auburn, AL

- Scraped a dermatology dataset of 5K+ labeled images, applying augmentation and ETL preprocessing to stabilize training.
- Uncovered misclassification biases linked to skin tone after evaluating EfficientNet-B0 with grouped confusion matrices.

Software Engineer Intern

May 2025 – Aug 2025

Kahua Inc. – SaaS for construction

Alpharetta, GA

- Decreased automated test development time by 30% by enabling code modification without resetting the test driver.
- Improved deployment reliability by expanding an Azure DevOps testing pipeline, automating 20+ key client workflows.

Founder

Jun 2021 – Dec 2024

Summer Sports Camp

Atlanta, GA

- Scaled to 100+ campers and 6 staff each summer, generating \$100K+ in profit before transferring ownership.
- Deployed a customer-response agent with AWS Pub/Sub, automating 75% of client inquiries and eliminating wait periods.

PROJECTS

AI Lifeguard System

Mar 2026

OpenCV, PyTorch

- Partnered with university facilities to build a pool-safety system integrating hardware, transport, and inference.
- Built custom hardware including a CAD camera ring and multiplexed video streams for wide-angle underwater capture.
- Shipped a lifeguard-facing GUI to stitch camera feeds, run real-time CV detection, and trigger drowning alerts.

Geopolitical Trend Forecaster

Oct 2025

Python, PyTorch, Pandas, Streamlit

[View Repository](#)

- Built an ETL pipeline from 200+ years of economic, demographic, military, and trade data from open-source archives.
- Developed quantitative scoring algorithms for country strength and geographic advantage to model competitiveness.
- Trained temporal convolutional networks to project geopolitical shifts and relative country standings 30 years forward.

Social Platform

Aug 2024

React, PostgreSQL, WebSockets, AWS, Docker

[View Repository](#)

- Built a calendar-sharing and messaging platform with real-time WebSocket updates and an AI-assisted scheduling agent.
- Developed a modular REST API with PostgreSQL supporting users, events, messaging, and agent-driven workflows.
- Deployed a containerized full stack on AWS EC2 using Docker Compose, Cloudflare DNS/TLS, and Caddy routing.

JupJewels.com

Jul 2025 – Sep 2025

React, Shopify, JavaScript

- Redesigned the storefront with the founder, increasing site traffic by 25% and improving conversion rates.
- Developed a custom jewelry builder that became the top-selling product, generating \$4K in revenue in 30 days.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, C#, HTML/CSS, SQL

Frameworks & Libraries: React, Flutter, PyTorch, Pandas, Selenium, NumPy, Playwright

Tools: AWS, Git, Azure, Azure DevOps, Node.js, Power BI, Bash, REST, Firebase, Docker, Robot Framework

Awards: ICPC Southeast Regional 3rd Place 2023 & 2025, GHSA AAA Regional 153 lb Wrestling Champion 2023